

C14502Q18

18. ① $H_0: p_1 = p_2 = p_3 = \frac{1}{3}$

$O_1 = 83 \quad O_2 = 61 \quad O_3 = 56$

$H_A: p_i \neq \frac{1}{3}$

$E_i = \frac{1}{3} \times 200 = 66.67$

② $\alpha = 0.05$

④ $\chi^2 = \frac{(83-66.67)^2}{66.67} + \frac{(61-66.67)^2}{66.67} + \frac{(56-66.67)^2}{66.67} = 6.19 > \chi^2_{2,0.05} = 5.991$

⑤ P.R. $\{X^2 | X^2 > \chi^2_{2,0.05}\}$

③ $X^2 = \sum_{i=1}^k \frac{(O_i - E_i)^2}{E_i} \sim \chi^2_{k-1, \alpha}$

⑥ $\therefore$ Reject H_0

95% CI: $\hat{p} \pm Z_{0.025} \times \sqrt{\frac{\hat{p}(1-\hat{p})}{n}} = 0.415 \pm 1.96 \times \sqrt{\frac{0.415 \times 0.585}{200}} = 0.415 \pm 0.0683 \rightarrow (0.3467, 0.4833)$

C14502Q22

22. ① $H_0: p_1 = p_2 = \dots = p_7 = \frac{1}{7}$

$E_i = \frac{1}{7} \times 100 = 14.29$

$H_A: p_i \neq \frac{1}{7}$

② $\alpha = 0.05$

③ $\chi^2 = \sum_{i=1}^k \frac{(O_i - E_i)^2}{E_i} \sim \chi^2_{k-1, \alpha}$

④ $\chi^2 = \frac{(24-14.29)^2}{14.29} + \frac{(36-14.29)^2}{14.29} + \frac{(27-14.29)^2}{14.29} + \frac{(26-14.29)^2}{14.29} + \frac{(32-14.29)^2}{14.29} + \frac{(26-14.29)^2}{14.29} + \frac{(29-14.29)^2}{14.29} = 3.629 < \chi^2_{6,0.05} = 12.592$

⑤ P.R. $\{X^2 | X^2 > \chi^2_{6,0.05} = 12.592\}$

⑥ $\therefore$ Non-reject H_0

C14503Q18

	Small	Medium	Large	Row total
Fatal	67	26	16	109
Not	128	63	46	237
Col. total	195	89	62	346

	Small	Medium	Large
Fatal	61.43	28.04	19.53
Not	133.57	60.96	42.47

① $H_0: \text{Size and fatality are independent}$

$H_A: \text{Size and fatality are dependent}$

② $\alpha = 0.05$

③ $\chi^2 = \sum_{i=1}^k \frac{(O_i - E_i)^2}{E_i} \sim \chi^2_{(r-1)(c-1), \alpha}$

④ $\chi^2 = \frac{(67-61.43)^2}{61.43} + \dots + \frac{(46-42.47)^2}{42.47} = 1.884 < \chi^2_{2,0.05} = 5.991$

⑤ P.R. $\{X^2 | X^2 > \chi^2_{2,0.05} = 5.991\}$

⑥ $\therefore$ Non-reject H_0

C14503023

No
Date

23. Eij:

	less than one	at least 1, not all	All at home
under 50/yr	36.267	21.000	10.653
50-74/yr	49.067	28.520	14.413
75, or 94/yr	35.200	20.460	10.340
Home type	39.467	22.940	11.593

Q. 1) H_0 : Factor A and B are independent
 H_A : Factor A and B are dependent

2) $\alpha = 0.05$

3) $\chi^2 = \frac{\sum (O_{ij} - E_{ij})^2}{E_{ij}} \sim \chi^2_{0.05, 6}$

4) $\chi^2 = 6.447 < 12.592$

5) R.R.: $\{ \chi^2 | \chi^2 > \chi^2_{0.05, 6} = 12.592 \}$

6) $\therefore$ Non-reject H_0

b. $p\text{-value} = P(\chi^2_{0.05, 6} > 6.447) > 0.1 \Rightarrow 0.1 > 0.05$
 $\therefore$ Non-reject H_0

C14504010

ID Edu	married			Eij	Yes	No
	Yes	No				
Bachelor or higher	187	113	300	Bachelor	166	134
some college	162	138	300	college	166	134
HS or less	149	151	300	HS	166	134
	498	402	900			

1) $H_0: p_1 = p_2 = p_3$

$H_A: \exists p_i \neq p_j \ (i \neq j)$

2) $\alpha = 0.01$

3) $\chi^2 = \sum \frac{(O_{ij} - \hat{E}_{ij})^2}{\hat{E}_{ij}} \sim \chi^2_{0.01, 2}$

4) $\chi^2 = \frac{(187-166)^2}{166} + \frac{(162-166)^2}{166} + \frac{(149-166)^2}{166} + \frac{(113-134)^2}{134} + \frac{(138-134)^2}{134} + \frac{(151-134)^2}{134}$
 $= 10.06 > \chi^2_{0.01, 2} = 9.21$

5) R.R.: $\{ \chi^2 | \chi^2 > \chi^2_{0.01, 2} = 9.21 \}$

6) Reject H_0

C14504 Q12

No
Date

12.	Apartment	Duplex	Single Residence	
1	8	20	1	29
2	16	8	9	33
3	10	10	14	34
4 or more	6	2	16	24
	40	40	40	120

$\bar{E}_{ij}$	Appt	Dup	Single
1	9.67	9.67	9.67
2	11	11	11
3	11.33	11.33	11.33
4	8	8	8

$$H_0: P_1 = P_2 = P_3$$

$$H_a: P_i \neq P_j \quad (i \neq j)$$

$$\alpha = 0.01$$

$$\chi^2 = \frac{\sum (O_{ij} - \bar{E}_{ij})^2}{\bar{E}_{ij}} \sim \chi^2_{0.01, 6}$$

$$\chi^2 = \frac{(8 - 9.67)^2}{9.67} + \dots + \frac{(16 - 8)^2}{8} = 36.4929 > \chi^2_{0.01, 6} = 16.8119$$

$$R.R.: \chi^2 > \chi^2_{0.01, 6} = 16.8119$$

∴ Reject H_0

C15501 Q12

12. RANK	1	2.5	2.5	5	5	5	7.5	7.5	9	10
Area	4.7	4.8	4.8	4.9	4.9	4.9	5.0	5.0	5.1	5.2
Sample	B	B	A	B	B	A	B	A	A	A

Q. ① H_0 : There is no difference in dissolved oxygen levels (Above = Below)
 H_a : Above > Below

$$\alpha = 0.05$$

$$T_{STAT} = \min(T_1, T_1^*) \sim T_{0.05}$$

$$T_1 = 2.5 + 5 + 7.5 + 9 + 10 = 34$$

$$T_1^* = 5(5+5+1) - 34 = 21$$

$$T^* = 21 > 19 = T_{0.05}$$

$$R.R.: T_1 > T_{0.05} = 19$$

∴ Non-reject H_0

b. Above: $\bar{X}_1 = 5$

$$s_1^2 = 0.025$$

Below: $\bar{X}_2 = 4.86$

$$s_2^2 = 0.03$$

$$s_p^2 = \frac{(n_1-1)s_1^2 + (n_2-1)s_2^2}{n_1+n_2-2} = 0.019$$

① $H_0: \mu_1 - \mu_2 = 0$, $H_a: \mu_1 - \mu_2 > 0$

② $\alpha = 0.05$

③ $t_{stat} = \frac{\bar{X}_1 - \bar{X}_2}{\sqrt{s_p^2 (\frac{1}{n_1} + \frac{1}{n_2})}} \sim t_{0.05, 8}$

④ $t^* = \frac{5 - 4.86}{\sqrt{0.019 (\frac{1}{5} + \frac{1}{5})}} = 1.606 < 1.860 = t_{0.05, 8}$

⑤ R.R: $\{t | t > t_{0.05, 8} = 1.860\}$

⑥ $\therefore$ Non-reject H_0

U5501 Q13

13. H_0 : Deaf = Hearing

H_a : Deaf $\neq$ Hearing

① $\alpha = 0.05$

② $T = \min(T_1, T_1^*)$

③ $T_1 = 45$

$$T_1^* = 9(9+9+1) - 45 = 126$$

$$T^* = \min(T_1, T_1^*) = 45 < T_{0.025} = 62$$

④ R.R: $\{T | T < T_{0.025} = 62\}$

⑤ $\therefore$ Reject H_0